Association Between Detention Status and Primary Multidrug-Resistant Tuberculosis Infection in Moldova

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Introduction

Tuberculosis (TB) is highly prevalent in former Soviet Union countries including Moldova (Brown et al., 2021). Multi-drug resistant tuberculosis (MDR-TB) is one of the most pressing public health issues currently and is highly prevalent in Moldova (En-Li Tan et al., 2025). Containing the spread of MDR-TB is of the utmost importance for objectives in global health, with worries that soon the rate of bacterial resistance might outpace the creation of new antibiotic drugs (En-Li Tan et al., 2025). TB is notoriously difficult to treat, requiring long antibiotics regimens that must be strictly adhered to, and if not, one risks developing MDR-TB (WHO, 2024). Previous research demonstrated that detention status was associated with increased MDR-TB risk and expanding on that knowledge will reveal important public health information (Jenkins et al., 2014). There is a research gap in understanding the relationship between primary MDR-TB contraction and detention status, as opposed to including re-infection cases as well. Investigating primary infection will reveal information on how people in detention facilities may be more likely exposed to MDR strains of TB on their initial infection, rather than developing it based on re-infection. This study aims to explore this association from all TB cases collected in Moldova over a roughly two-year period.

This investigation aimed to determine whether having been in detention is associated with the development of primary multi-drug resistant tuberculosis (MDR-TB) infection, among TB positive adults in Moldova from 2008-2010. Additionally, the effect of HIV coinfection as an effect modifier on this relationship was investigated.

It is hypothesized that detention status is positively associated with higher rates of contracting MDR-TB, as it is known that TB rates in detention facilities are higher than the average exposure. Additionally, it is hypothesized that HIV coinfection would increase the association between detention status and MDR-TB, so that the OR is higher among those with HIV coinfection compared to those without HIV coinfection.

Methods

Data Source and Study Population

The data used for this analysis was routinely collected surveillance data of all TB cases reported in Moldova from January 1st, 2008, to December 31st, 2010. Data was collected through an online database that collected laboratory results, treatment outcomes, and demographic data at initial diagnosis. Data collection is managed through the National Center of Health Management and National Tuberculosis Program.

The dataset was restricted to only new cases, meaning TB cases reported as the first time that subject contracted TB. This decision was made because the study aims to find whether detention status is associated with contracting MDR-TB from another host. MDR-TB can be developed from an initial non-MDR-TB case, which should have little association with detention status and could be an alternative causal pathway. Restricting the dataset in this manner ensures that all MDR-TB cases identified were most likely contracted from others.

Outcome and Exposure Definition

MDR-TB case outcome was defined as if the participant's laboratory results showed they were resistant to both Rifampin and Isoniazid. Resistance to Rifampin and Isoniazid are the standard way of classifying a TB case as MDR, as those are the primary treatment drugs for non-MDR-TB. If participants had only one of the two resistances, they were categorized as not having MDR-TB. The exposure of detention status was defined through the question of if the participant has ever been in detention on the demographics questionnaire completed at the time of diagnosis. HIV coinfection status was defined as those who were tested and positive for HIV at the time of their TB diagnosis. Those who were tested with unknown results were marked as missing for HIV coinfection. This decision was made as when the test is marked as unknown, there is a large amount of uncertainty as to which status they would be, which could affect the analysis if mislabeled, and so marking as missing in the dataset is the most conservative way to not bias the data.

Covariate Selection

Covariates identified in this investigation were sex, rural status, homelessness status, employment status, education status, and HIV coinfection. Covariates were identified via a Directed Acyclic Graph (DAG). All covariates were categorized based on subjects' answers to the demographics questionnaire. Employment status was categorized into working, disabled, pensioner, student, or unemployed. Education status was categorized into none, primary, secondary, specialized secondary, or higher education based on self-reported questionnaire data. Data were excluded if they did not have valid data for detention status or MDR-TB status.

Statistical Analysis

Participant characteristics were described using summary statistical measures. Categorical variables (sex, urban/rural status, education status, employment status, homelessness, HIV status) were described using frequency and percentage. Continuous variables (age) were described using mean and standard deviation. The distribution of variables was stratified by detention status. Differences between groups were assessed using chi squared tests for categorical variables and two sample T tests for continuous variables. Additionally, missingness of variables was investigated and noted.

Multivariable logistic regression was used to examine the association between detention status and MDR-TB status. A crude model was fitted between the main exposure and outcome variables. An adjusted model was constructed, controlling for all covariates identified as potential confounders of the relationship between the main outcome and exposure (age, sex, urban/rural status, education status, employment status, homelessness, HIV status). Confounder selection was theory driven through a directed acyclic graph. If the adjusted OR differed from the crude OR by 10% or more, covariate adjustments were retained. Results are presented as odds ratios with 95% confidence intervals and p values for both models.

The interaction of HIV coinfection was investigated through introducing HIV coinfection as an interaction term to the previous model. Statistical evidence of HIV coinfection as an effect modifier was assessed using a Wald test on the interaction term in the regression model.

Due to substantial missingness of data associated with HIV coinfection, a sensitivity analysis was conducted. First, HIV coinfection was removed from the adjusted model to determine any effects on the main association. Second, a multiple imputation was conducted to impute missing values of HIV coinfection status. Assuming that data was missing at random, all variables in the dataset were used as predictors in the imputation model, and six imputed datasets were generated. The results were then pooled across all six imputations, and OR and 95% CI reported. The results of the sensitivity analysis are visualized in a forest plot.

All analyses were conducted using R version 4.3.2

Results

A total of 7966 individuals with newly diagnosed tuberculosis were identified in Moldova between January 2008 and December 2010. Following the exclusion of individuals with missing detention status or MDR-TB classification, 7792 participants were included in the final analytic sample. Of this sample, 8.2% were classified as having MDR-TB and 9.5% reported ever having been detained. The baseline characteristics of the cohort stratified by detention status are reported in Table 1 below.

Table 1. Baseline Characteristics of TB Cases in Moldova, Stratified by Detention Status, 2008-2010

Characteristic	**Never In Detention** N = 7,050[1]	**In Detention at Any Time** N = 742[1]	**p-value**[2]
Sex			<0.001
Female	2,307 (33%)	41 (5.5%)	
Male	4,743 (67%)	701 (94%)	
Age at Diagnosis	39 (16)	34 (11)	<0.001
Urban/Rural Residence			<0.001
Urban	3,106 (44%)	443 (60%)	
Rural	3,944 (56%)	299 (40%)	
Homelessness Status			0.045

Not Homeless	6,590 (96%)	685 (94%)	
Homeless	296 (4.3%)	43 (5.9%)	
Missing	164	14	
Occupational Status			<0.001
Worker	1,503 (21%)	56 (7.6%)	
Disabled	425 (6.1%)	18 (2.4%)	
Pensioner	576 (8.2%)	12 (1.6%)	
Student	425 (6.1%)	28 (3.8%)	
Unemployed	4,065 (58%)	627 (85%)	
Missing	56	1	
Educational Status			<0.001
None	239 (3.4%)	9 (1.2%)	
Primary	1,558 (22%)	147 (20%)	
Secondary	3,811 (55%)	465 (63%)	
Specialized Secondary	1,061 (15%)	109 (15%)	
Higher	323 (4.6%)	9 (1.2%)	
Missing	58	3	
HIV Coinfection Status			<0.001
No HIV Co-Infection	5,899 (96%)	559 (85%)	
HIV Co-Infection	232 (3.8%)	101 (15%)	
Missing	919	82	
[1]n (%); Mean (SD)			
[2]Pearson's Chi-squared test; Wilcoxon rank sum test			

The mean age at diagnosis was 39 (SD: 16) among non-detained individuals and 34 (SD: 11) among detained individuals. Detained individuals were more likely to be male compared to those never detained (94% vs. 67%, p <0.001). Detained participants were more likely to be unemployed compared to non-detained participants (85% vs. 58%, p <0.001) and less likely to be workers (8% vs. 21%). Detained participants were more likely to reside in urban areas compared to non-detained participants (60% vs. 44%, p <0.001). Detained participants had a higher rate of HIV coinfection compared to non-detained participants (15% vs. 4%, p <0.001).

There was a substantial amount of missingness in regard to the HIV coinfection variable. 919 (13%) of non-detained participants had missing values for HIV coinfection status, while 82 (11%)

were missing among detained participants. Additionally, homelessness had missingness, with both detention and non-detention participants having approximately 2% missing homelessness status values. Occupational and Educational status had a low amount of missingness.

In a crude analysis, individuals who had been detained at any time had 79.4% higher odds of contracting MDR-TB compared to those who had not been detained at any time (OR: 1.794, 95%CI: 1.42, 2.25, $p < 0.001$).

After adjusting for potential confounders, the OR was substantially attenuated (OR: 1.20, 95%CI: 0.91, 1.56, $p = 0.17$). The adjusted OR differed from the crude OR by approximately 20% and so the adjusted model was retained for further analyses. However, the association between detention status and MDR-TB was no longer significant, in the adjusted analyses. A model testing the association between detention status and MDR-TB, adjusted for confounding, with HIV coinfection as an interaction term, found that the interaction term between HIV coinfection and detention status was not statistically significant (Wald Test, p=0.25).

Table 2. Crude and Adjusted Odds Ratios for the Association Between Detention Status and MDR-TB, 2008-2010

	Unadjusted			Adjusted		
Characteristic	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
Detention Status						
Never In Detention	Ref.	Ref.		Ref.	Ref.	
In Detention at Any Time	1.79	1.42, 2.25	<0.001	1.20	0.92, 1.56	0.20
Sex						
Female				Ref.	Ref.	
Male				1.43	1.16, 1.78	<0.001
Urban/Rural Residence						
Urban				Ref.	Ref.	
Rural				0.84	0.70, 1.01	0.070
Homelessness Status						
Not Homeless				Ref.	Ref.	
Homeless				1.39	0.94, 1.98	0.082
Occupational Status						
Worker				Ref.	Ref.	

Disabled				0.58	0.32, 0.98	0.055
Pensioner				1.68	0.93, 2.88	0.071
Student				1.08	0.69, 1.66	0.70
Unemployed				1.23	0.96, 1.57	0.10
Educational Status						
None				Ref.	Ref.	
Primary				1.67	0.92, 3.29	0.11
Secondary				1.72	0.97, 3.34	0.083
Specialized Secondary				1.84	1.00, 3.66	0.064
Higher				1.39	0.66, 3.06	0.40
Age Category						
<25				Ref.	Ref.	
25-44				0.72	0.56, 0.91	0.007
44-65				0.60	0.46, 0.79	<0.001
65+				0.16	0.07, 0.38	<0.001
HIV Coinfection Status						
No HIV Co-Infection				Ref.	Ref.	
HIV Co-Infection				1.86	1.33, 2.56	<0.001
Abbreviations: CI = Confidence Interval, OR = Odds Ratio						

Notably, the adjusted model reduced the analytical sample by 1209 observations, coming primarily from the HIV coinfection status category. A sensitivity analysis was conducted, removing HIV coinfection from the adjusted model, which resulted in an OR of 1.36 (95%CI: 1.07, 1.74, $p = 0.013$). While the official OR of this investigation will be considered non-significant, the missingness of HIV coinfection may be an explanation. Additionally, when a multiple imputation was conducted on the data set, the pooled OR found was 1.27 (95%CI: 0.99, 1.63, $p = 0.056$), but it was not statistically significant. Sensitivity analyses results comparing the crude, adjusted, HIV-removed, and imputation models are presented in a forest plot below (OR, 95%CI).

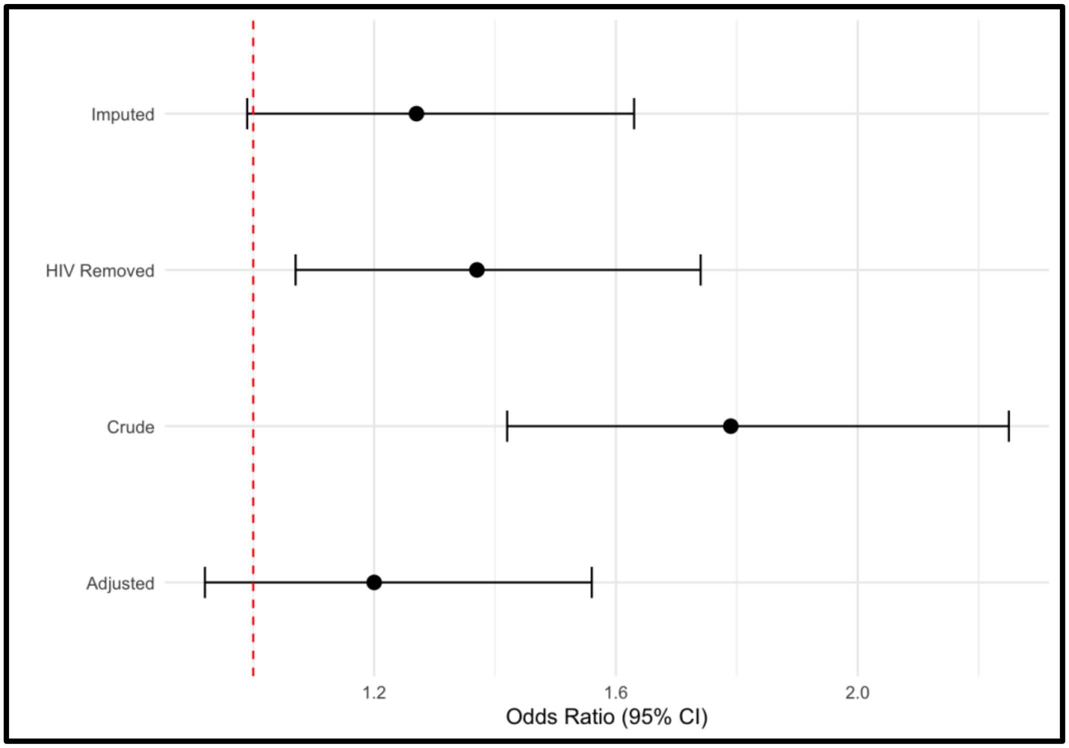

Figure 1. Forest plot of crude, adjusted, HIV-removed, and multiple imputation models for the association between detention status and MDR-TB, Moldova 2008-2010 (OR, 95% CI)

The primary model demonstrated moderate discrimination (C-statistic: 0.62). The Hosmer-Lemeshow test indicated adequate model fit ($\chi^2(8) = 8.75$, $p = 0.36$).

Discussion

This investigation found no statistically significant association between detention status and MDR-TB, after controlling for confounding. This indicates that there may be alternative explanations for a higher rate of MDR-TB among those who have been detained, suggesting potential mediation effects or other primary causal pathways. A sensitivity analysis revealed that this association could be clouded in the sample by the extent of missingness in the data of HIV coinfection.

Places of detention are environments with high population density and limited resources, conditions which are conducive to high TB transmission. Existing reservoirs of MDR-TB can spread more readily through primary transmission, rather than requiring re-infection and acquired resistance like in the traditional pathway. The role of HIV coinfection in this relationship may require further discussion. While HIV coinfection was not found to be a significant effect modifier, substantial missingness in HIV coinfection status reduced the statistical power of the analysis. When HIV coinfection status was removed from the model, the association between MDR-TB and detention status became statistically significant, and when missing HIV status was imputed, the association was borderline significant. This suggests that the non-significant finding in the primary model may be at least partially explained by missingness of HIV coinfection status.

This investigation has several strengths. The restriction to only include newly diagnosed MDR-TB cases strengthens the causal inference by closing alternative biological pathways. Covariate selection was theory driven from a DAG, which reduced the risk of any ad-hoc adjustment. There were, however, several limitations that should be noted. The largest limitation was the substantial missingness in the HIV coinfection covariate, which reduced the study sample by roughly 1200 observations. The binary detention variable may not be the most accurate method for measuring detention status, as it captures a wide variety of different detention experiences which may not directly speak to the causal pathway of contracting MDR-TB due to being in a detention facility. Homelessness and occupation could potentially lie on a causal pathway between detention status and MDR-TB, as detention could lead to homelessness and loss of occupation leading to MDR-TB. If this were the case, the model may underestimate the true association. Despite the non-significant primary finding, detention facilities remain plausible sites of high MDR-TB transmission.

This investigation has implications for public health. While a clear link could not be determined, it is known that detention facilities are likely places of high transmission of MDR-TB (Nyasulu et. al., 2025). Improving the quality of these facilities in terms of hygiene and medical care is an important measure to be taken to reduce the transmission of MDR-TB. Fighting MDR-TB on all fronts, both through primary transmission and reducing acquired resistance, is of the utmost importance for public health. The rate of drug resistance is in a constant battle with the speed at which new drugs can be developed.

Conclusion

The investigation found no statistically significant association between detention status and multi-drug resistant tuberculosis, among first case transmission in Moldova from 2008-2010. However, a sensitivity analysis revealed that missingness of HIV coinfection status may have attenuated the observed relationship, and a true relationship cannot be ruled out.

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